

```
from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

```
!nvidia-smi
```

Mon Oct 13 16:18:25 2025

```
+-----+
+-----+
| NVIDIA-SMI 550.54.15                  Driver Version: 550.54.15
| CUDA Version: 12.4                  |
+-----+-----+
+-----+-----+
| GPU   Name                               Persistence-M | Bus-Id        Disp.A |
| Volatile Uncorr. ECC |
| Fan  Temp  Perf          Pwr:Usage/Cap |      Memory-Usage |
| GPU-Util  Compute M. |
| MIG M. |
|
+-----+-----+
+-----+-----+
|    0   Tesla T4                                  Off |   00000000:00:04.0 Off |
| 0 |
| N/A     52C    P8              10W /   70W |      0MiB /  15360MiB |
| 0%      Default |
|
| N/A |
+-----+-----+
+-----+-----+

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+-----+-----+
| Processes:
|
| GPU   GI    CI          PID    Type    Process name
GPU Memory |
|      ID    ID
Usage      |
|
+-----+-----+
+-----+-----+
| No running processes found
|
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```

```

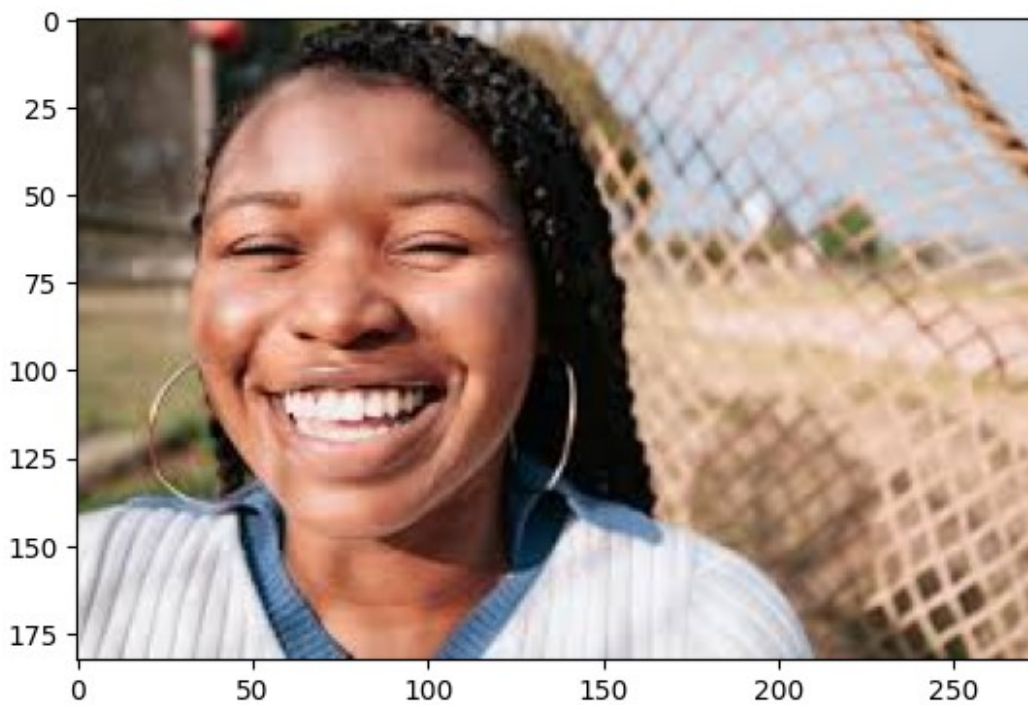
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.preprocessing import image
import matplotlib.pyplot as plt
import tensorflow as tf
import numpy as np
import cv2
import os

img = image.load_img('/content/drive/MyDrive/Colab Notes/CNN-Happy or
Sad/training/happy/2.jpeg')

plt.imshow(img)

<matplotlib.image.AxesImage at 0x7d057588c710>

```



```

i1 = cv2.imread('/content/drive/MyDrive/Colab Notes/CNN-Happy or
Sad/training/happy/2.jpeg')
i1
array([[ 52,  66,  72],
       [ 48,  62,  68],
       [ 42,  56,  62],
       ...,
       [209, 195, 183],
       [200, 186, 174],
       [186, 172, 160]],
      dtype=uint8)

```

```

[[ 48, 62, 68],
 [ 45, 59, 65],
 [ 41, 55, 61],
 ...,
 [222, 208, 196],
 [216, 202, 190],
 [206, 192, 180]],

[[ 43, 57, 63],
 [ 41, 55, 61],
 [ 39, 53, 59],
 ...,
 [229, 215, 203],
 [226, 212, 200],
 [222, 208, 196]],

...,

[[206, 203, 205],
 [250, 247, 249],
 [236, 233, 235],
 ...,
 [133, 159, 196],
 [150, 174, 210],
 [167, 191, 227]],

[[213, 210, 212],
 [253, 250, 252],
 [237, 234, 236],
 ...,
 [157, 180, 218],
 [169, 191, 227],
 [181, 200, 237]],

[[224, 221, 223],
 [251, 248, 250],
 [238, 235, 237],
 ...,
 [174, 196, 232],
 [178, 198, 233],
 [181, 200, 235]]], dtype=uint8)

```

```
i1.shape
```

```
# height, weight, rgb of the image
```

```
(183, 275, 3)
```

```
train = ImageDataGenerator(rescale=1/255)
```

```
validation = ImageDataGenerator(rescale=1/255)
```

```

train_dataset =
train.flow_from_directory('/content/drive/MyDrive/Colab Notes/CNN-
Happy or Sad/training',
                           target_size=(200,200),
                           batch_size=32,
                           class_mode='binary')

validation_dataset =
validation.flow_from_directory('/content/drive/MyDrive/Colab
Notes/CNN-Happy or Sad/validation',
                              target_size=(200,200),
                              batch_size=32,
                              class_mode='binary')

Found 11 images belonging to 2 classes.
Found 0 images belonging to 2 classes.

train_dataset.class_indices
{'happy': 0, 'not happy': 1}

train_dataset.classes
array([0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1], dtype=int32)

# now we are applying maxpooling

model = tf.keras.models.Sequential([ tf.keras.layers.Conv2D(16,
(3,3),activation = 'relu',input_shape = (200,200,3)),
                                   tf.keras.layers.MaxPool2D(2,2), #3
# filter we applied here
                                   #
                                   tf.keras.layers.Conv2D(32,
(3,3),activation = 'relu'),
                                   tf.keras.layers.MaxPool2D(2,2),
                                   #
                                   tf.keras.layers.Conv2D(64,
(3,3),activation = 'relu'),
                                   tf.keras.layers.MaxPool2D(2,2),
                                   ##
                                   tf.keras.layers.Flatten(),
                                   ##
                                   tf.keras.layers.Dense(512,
activation = 'relu'),
                                   #
                                   tf.keras.layers.Dense(1,activation= 'sigmoid')
                                   ]
                                   )

/usr/local/lib/python3.12/dist-packages/keras/src/layers/
convolutional/base_conv.py:113: UserWarning: Do not pass an

```

`input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer,  
**kwargs)
```

```
model.compile(loss='binary_crossentropy',  
              optimizer = tf.keras.optimizers.RMSprop(learning_rate=  
0.001),  
              metrics = ['accuracy']  
)
```

```
model_fit = model.fit(train_dataset,  
                      epochs = 15)
```

Epoch 1/15

1/1 _____ 0s 80ms/step - accuracy: 0.7273 - loss: 0.4057

Epoch 2/15

1/1 _____ 0s 91ms/step - accuracy: 0.8182 - loss: 0.3034

Epoch 3/15

1/1 _____ 0s 97ms/step - accuracy: 1.0000 - loss: 0.2031

Epoch 4/15

1/1 _____ 0s 90ms/step - accuracy: 1.0000 - loss: 0.1558

Epoch 5/15

1/1 _____ 0s 88ms/step - accuracy: 1.0000 - loss: 0.1444

Epoch 6/15

1/1 _____ 0s 81ms/step - accuracy: 1.0000 - loss: 0.1414

Epoch 7/15

1/1 _____ 0s 143ms/step - accuracy: 1.0000 - loss: 0.1410

Epoch 8/15

1/1 _____ 0s 140ms/step - accuracy: 1.0000 - loss: 0.0881

Epoch 9/15

1/1 _____ 0s 101ms/step - accuracy: 1.0000 - loss: 0.0580

Epoch 10/15

1/1 _____ 0s 85ms/step - accuracy: 1.0000 - loss: 0.0347

Epoch 11/15

1/1 _____ 0s 103ms/step - accuracy: 1.0000 - loss: 0.0278

Epoch 12/15

1/1 _____ 0s 102ms/step - accuracy: 1.0000 - loss:

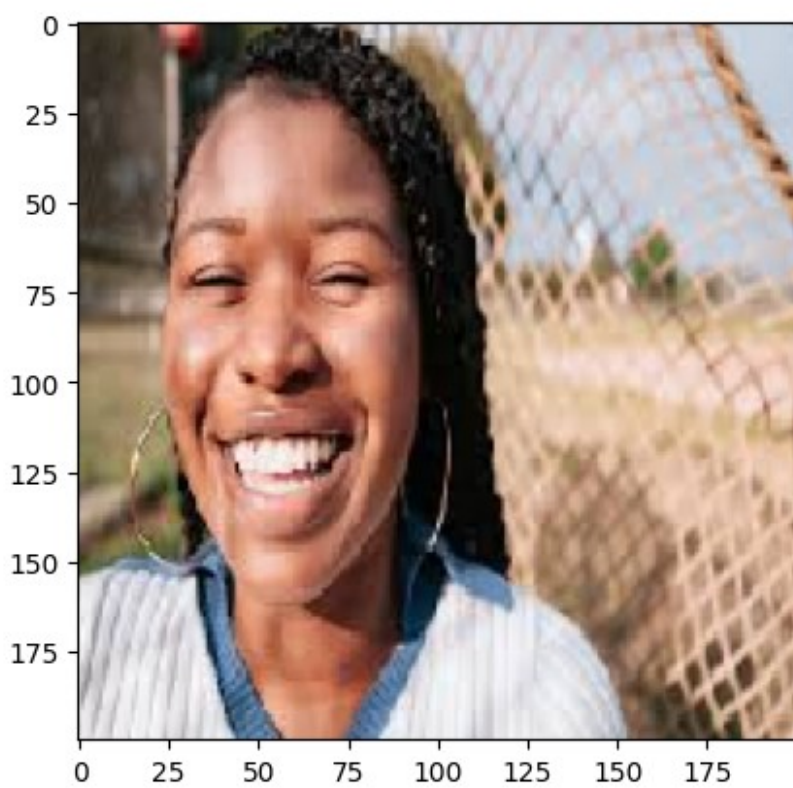
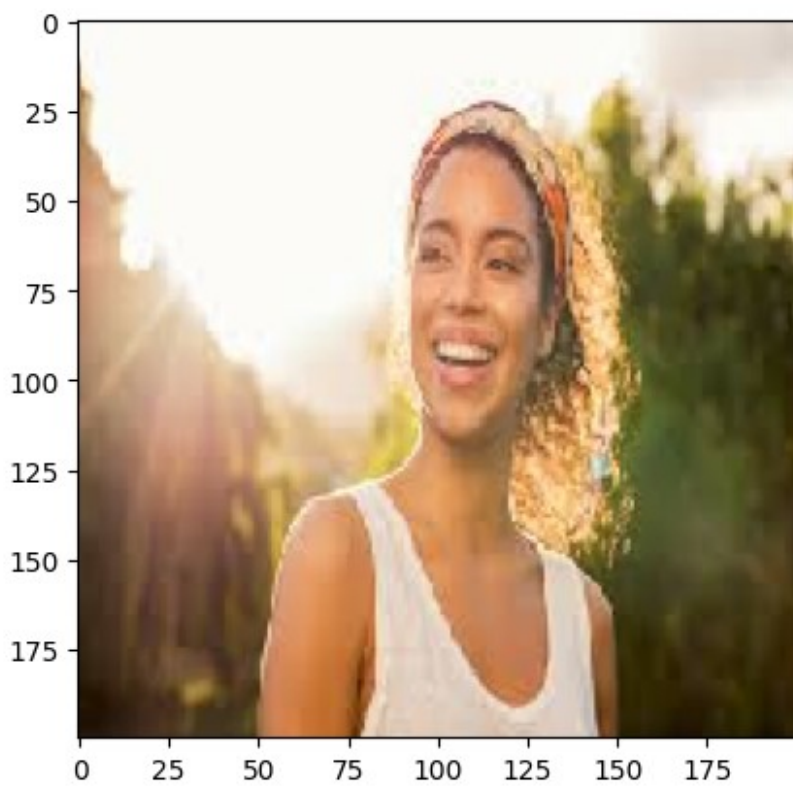
```
0.0232
Epoch 13/15
1/1 _____ 0s 96ms/step - accuracy: 1.0000 - loss:
0.0197
Epoch 14/15
1/1 _____ 0s 88ms/step - accuracy: 1.0000 - loss:
0.0169
Epoch 15/15
1/1 _____ 0s 91ms/step - accuracy: 1.0000 - loss:
0.0148
```

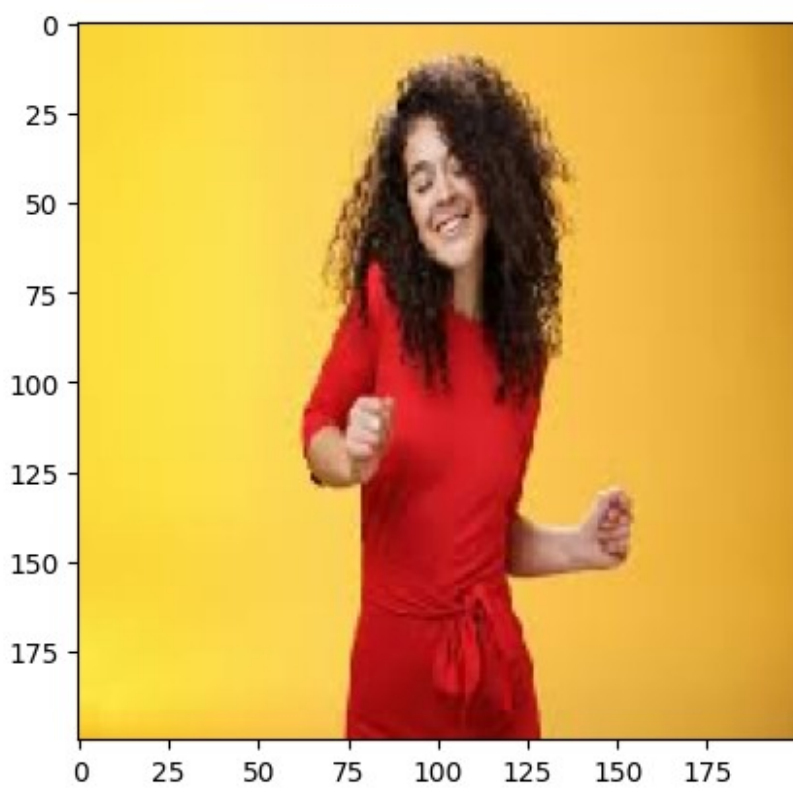
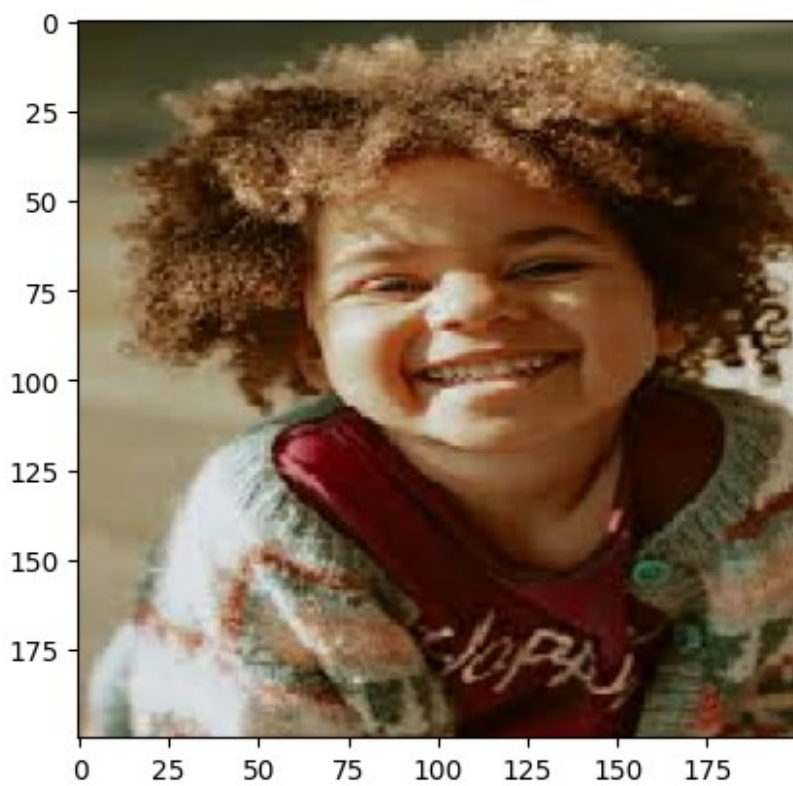
```
dir_path = '/content/drive/MyDrive/Colab Notes/CNN-Happy or
Sad/testing'
for i in os.listdir(dir_path):
    print(i)

1.jpeg
2.jpeg
3.jpeg
4.jpeg
5.jpeg
1.1.jpeg
2.1.jpeg
3.1.jpeg
4.1.jpeg
5.1.jpeg

dir_path = '/content/drive/MyDrive/Colab Notes/CNN-Happy or
Sad/testing'

for i in os.listdir(dir_path):
    img = image.load_img(dir_path+'/'+i,target_size=(200,200))
    plt.imshow(img)
    plt.show()
```











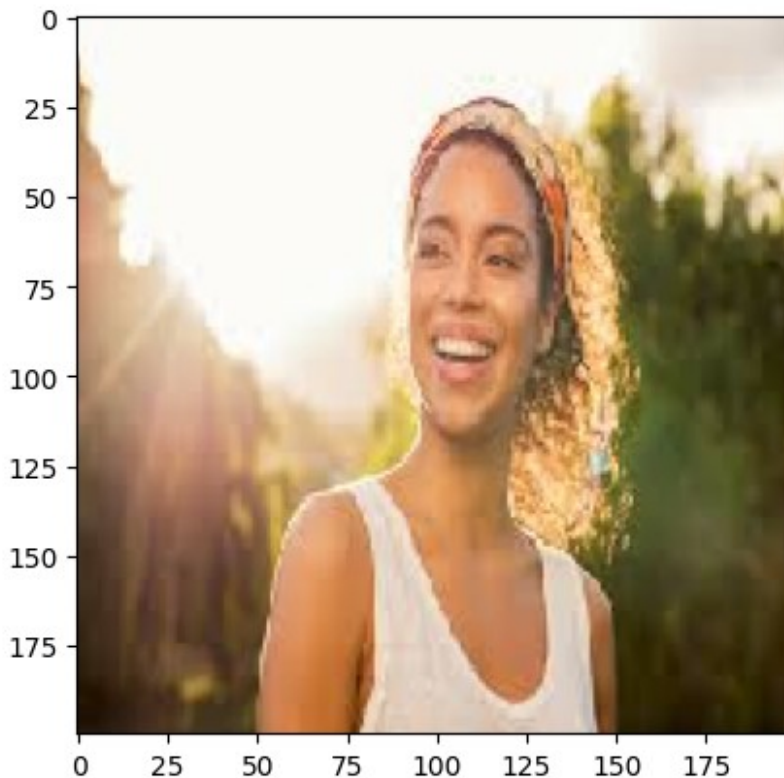
```

dir_path = r'/content/drive/MyDrive/Colab Notes/CNN-Happy or
Sad/testing'
for i in os.listdir(dir_path ):
    img = image.load_img(dir_path+ '//' +i, target_size = (200,200))
    plt.imshow(img)
    plt.show()

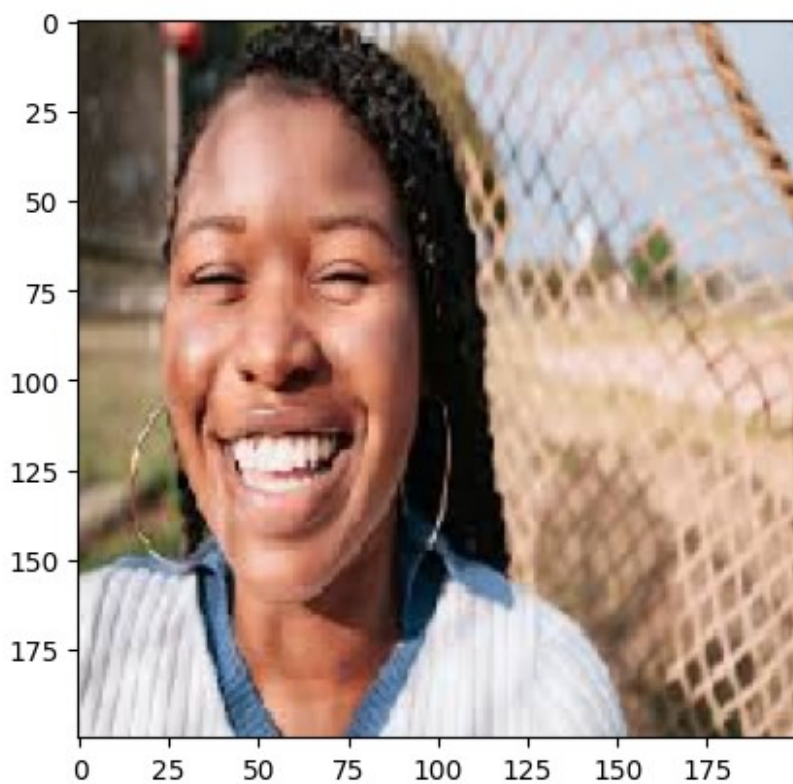
    x= image.img_to_array(img)
    x=np.expand_dims(x,axis = 0)
    images = np.vstack([x])

    val = model.predict(images)
    if val == 0:
        print( ' i am not happy')
    else:
        print('i am happy')

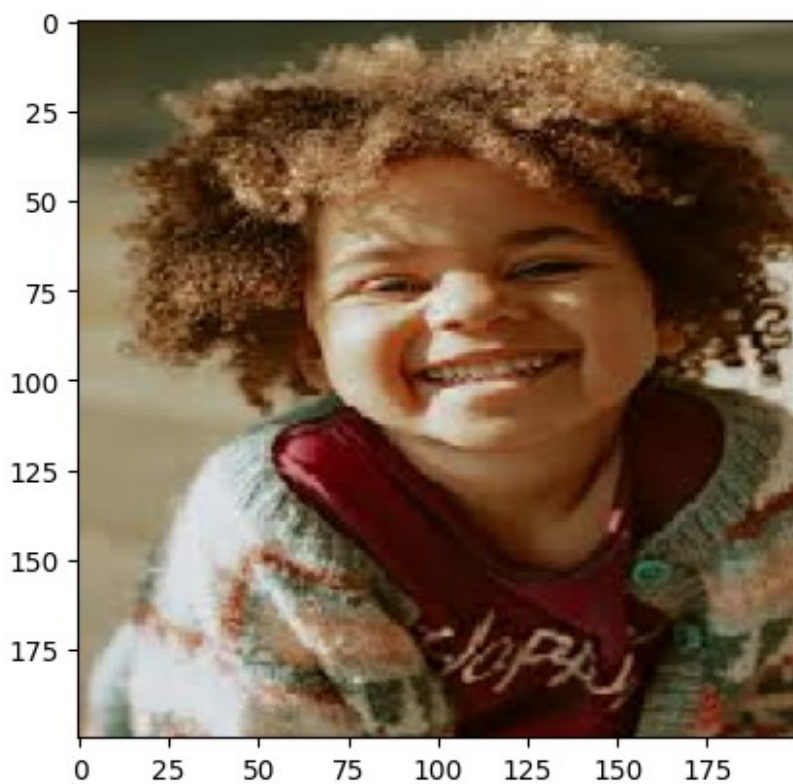
```



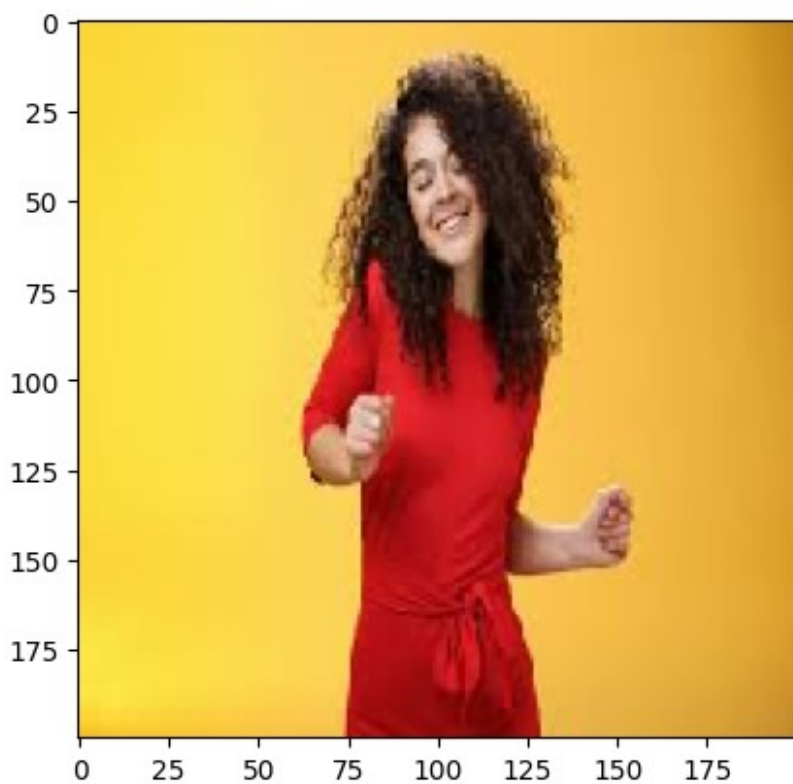
1/1 ————— 1s 836ms/step
i am not happy



1/1 ————— 0s 33ms/step
i am not happy



1/1 ————— 0s 33ms/step
i am not happy



1/1 ————— 0s 34ms/step
i am not happy



1/1 ————— 0s 36ms/step
i am not happy



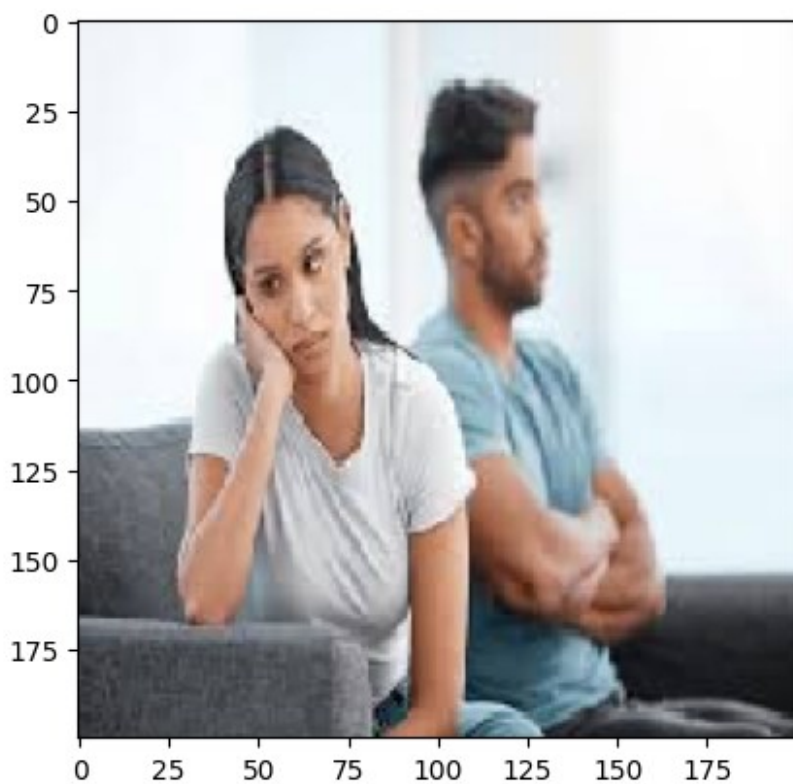
1/1 ————— 0s 34ms/step
i am happy



1/1 ————— 0s 40ms/step
i am happy



1/1 ————— 0s 36ms/step
i am happy



1/1 ————— 0s 40ms/step
i am happy



1/1 _____ 0s 41ms/step

i am happy

!nvidia-smi

Mon Oct 13 16:41:10 2025

```
+-----+
+-----+
| NVIDIA-SMI 550.54.15                  Driver Version: 550.54.15
| CUDA Version: 12.4                  |
+-----+-----+
+-----+-----+
| GPU   Name                               Persistence-M | Bus-Id        Disp.A | |
| Volatile Uncorr. ECC |                               |                  |
| Fan  Temp  Perf    Pwr:Usage/Cap       |      Memory-Usage |
| GPU-Util  Compute M. |                               |                  |
|                               | MIG M.             |                  |
|                               |                               |                  |
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=====+=====
=====|
|  0  Tesla T4                               Off |  00000000:00:04.0 Off |
0 |
| N/A    72C    P0              33W /   70W |  1146MiB /  15360MiB |
0%      Default |
```

```
|
| N/A |
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```

```
| Processes:
|
| GPU   GI   CI   PID   Type   Process name
GPU Memory |
|         ID   ID
Usage      |
|
```

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|-----+-----|
```